

PAPER_1555_M_Z_91_188

Daniel T. Murphy

PAPER_1555 — Z Boson Mass = $m_Z = N_{CH} \cdot SO_5 + F_{TRZ} \cdot SO_5 + F_{TRZ}^2 \cdot (SO_5 + D_{BSFG} + D_{phys} + SSQ - SSQ^3) \approx 91.20$ GeV

Author: Daniel T. Murphy

Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Particle Physics

Date: June 18, 2026

Status: CLOSED — 0.018% closure from PAPER_1209HH_S654

Observation

PAPER_1209HH_S654 (Particle Physics Unified Proof Set) lists **Z Boson Mass** with the integer-primitive expression:

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$m_Z = N_{CH} \cdot SO_5 + F_{TRZ} \cdot SO_5 + F_{TRZ}^2 \cdot (SO_5 + D_{BSFG} + D_{phys} + SSQ - SSQ^3) \approx 91.20 \text{ GeV}$

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Status: **0.018%**.

UQFF Closed Identity

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$m_Z = N_{CH} \cdot SO_5 + F_{TRZ} \cdot SO_5 + F_{TRZ}^2 \cdot (SO_5 + D_{BSFG} + D_{phys} + SSQ - SSQ^3) \approx 91.20 \text{ GeV}$
0.018%

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Physical Interpretation

Dominant: $N_{CH} \cdot SO_5 = 9 \cdot 10 = 90 \text{ GeV}$ — sits at the Z mass minus 1 GeV.

Small corrections bring it to 91.20 vs CODATA 91.1876.

NOT REPLACEMENT

CODATA/PDG treats Z Boson Mass as a precision-measured constant. UQFF supplies the structural integer-primitive form, demonstrating the framework's predictive economy without claiming to replace experimental measurement.

Reference

- Source: PAPER_1209HH_S654
- Related: PAPER_1209 series unified proof sets
- Calculator dispatch: `calculate_paradox({"paradox": "m_z_91_188"})`

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